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SERVER:
OPEN CHIPLET
ECONOMY



System level standards for Multi-vendor Chiplet Interoperability

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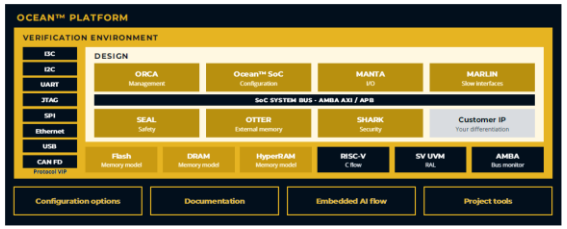
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GSOC.solutions

FAST FORWARD YOUR ASIC DEVELOPMENT

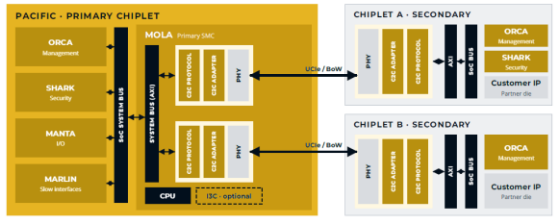
Ocean™ SOC platform

- RISC-V , Linux capable
- Design & Verification Duo
- Out of the BOX SOC project
- AI envelop for easy customization



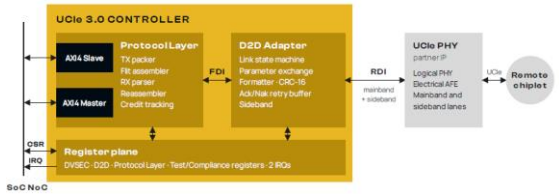
Ocean™ Chiplet SOC family

- Chiplet System management
- Optimized for Physical AI market
- Delivered as a complete project
- Plug selected PHY and go !



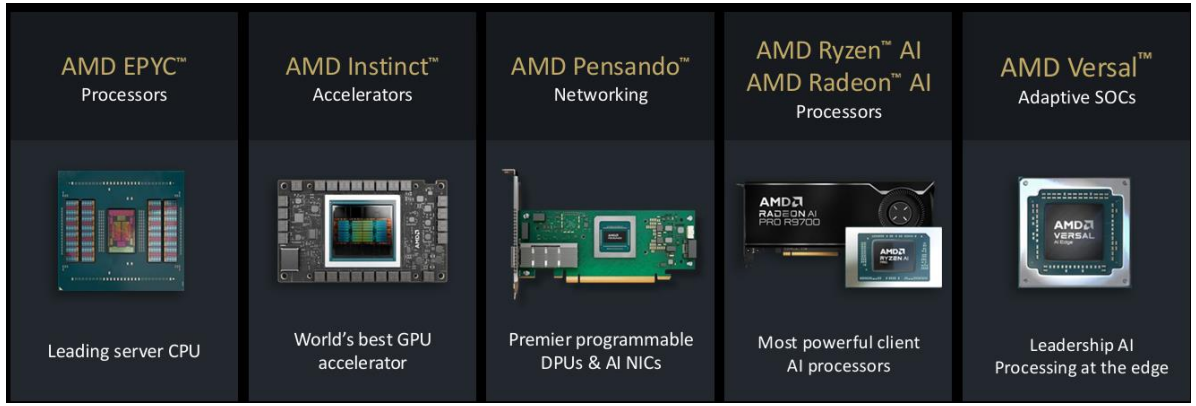
UCIe 3.0 Controller

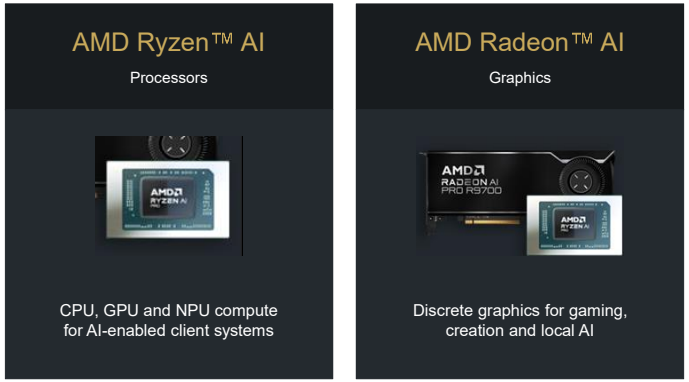
- Native AXI4 , for seamless c2c connection
- Protocol & Adapter layers
- Plug the PHY on the RDI



AMD

Best End-to-End AI Compute Portfolio in the Industry





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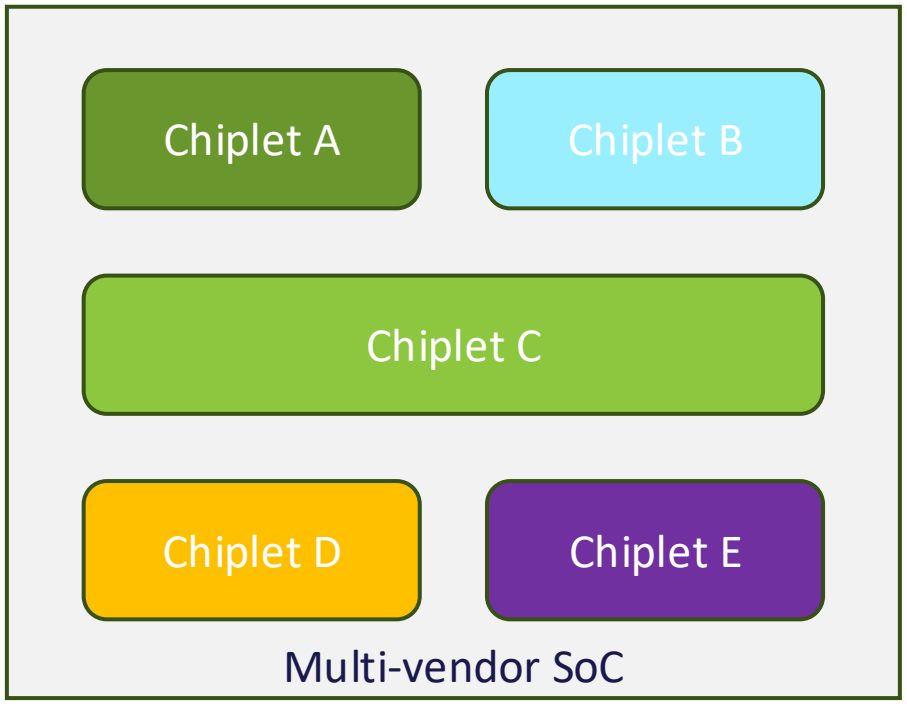
Background

- Chiplet interoperability is happening now
- OCP Bunch of Wires (BoW) and UCIe enable multi-vendor interoperability at the physical and protocol layers
- True multi-vendor interoperability requires alignment beyond those layers
- OCP Foundation Chiplet System Architecture (FCSA) is addressing broader interoperability for chiplet-based SoCs
- The OCP Chiplet Systems Workstream collaborates with FCSA, with a current focus on firmware APIs and message transport

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Specialized chiplets need shared services

Vendors differentiate through unique capabilities, but every chiplet must support common functions for the SoC to operate as one system.



Shared SoC services

- Power and performance
- Secure boot
- RAS detection and reporting
- Clock, voltage, and reset

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OCP Chiplet Firmware Stack

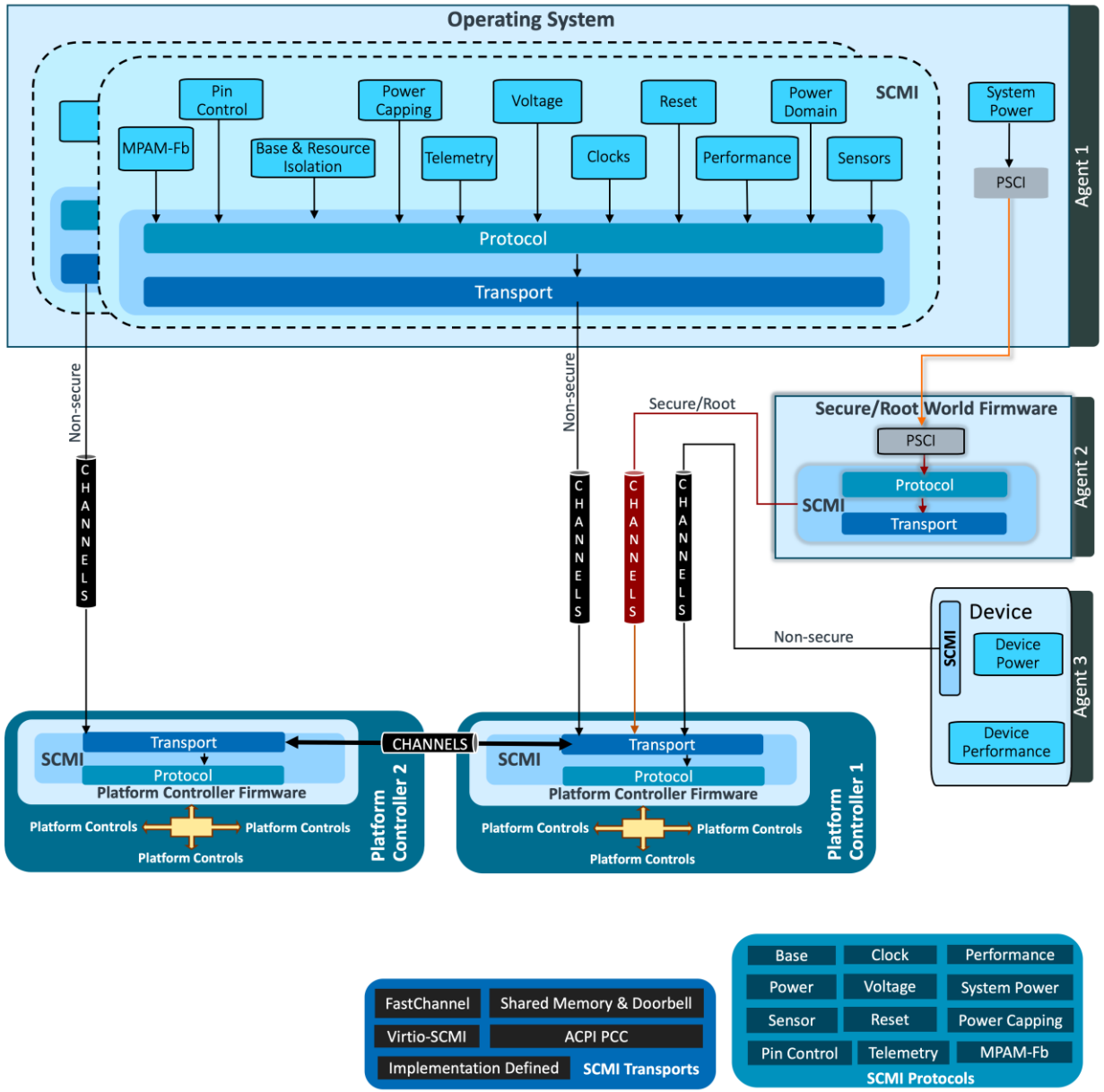
OCP Chiplet Systems focuses on the API, protocol, and transport layers; FCSA defines the physical layer.



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FW API and Protocol

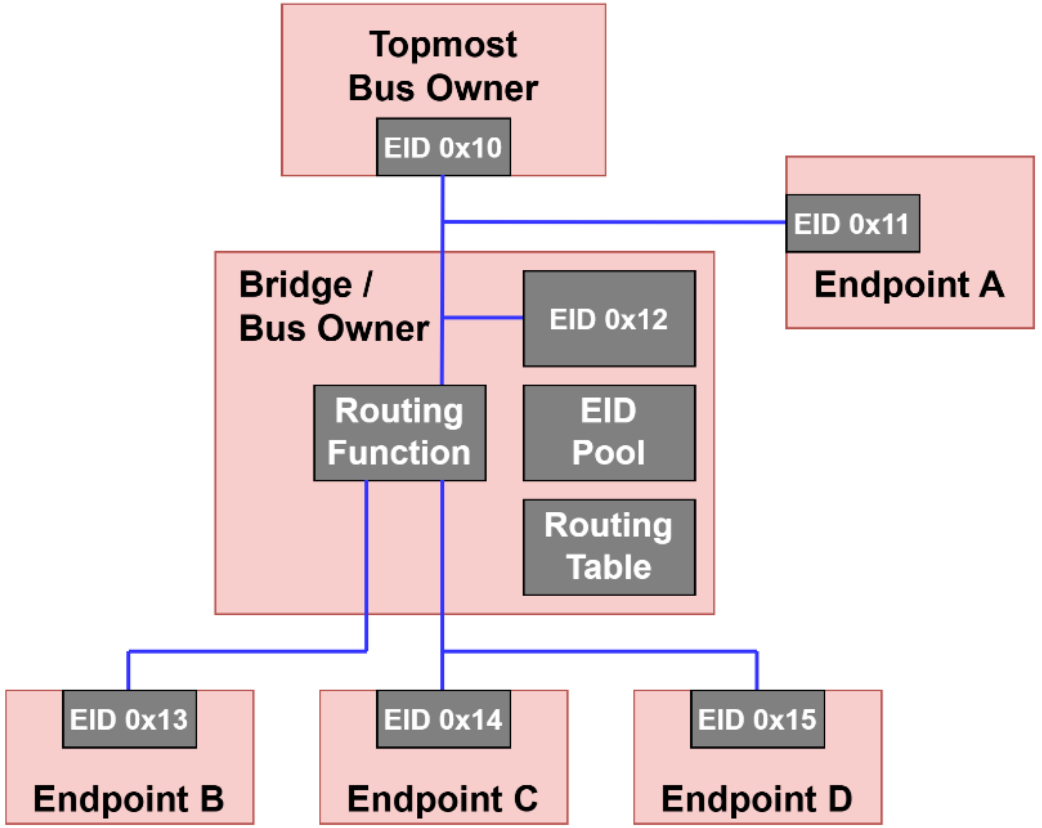
- Agent: An independent SCMI client—software or firmware, at any privilege level—that communicates with the platform independently.
- Message: A communication exchanged between an agent and the platform over an SCMI channel.
- Channel: The agent–platform transport path: either a dedicated SCMI FastChannel or a standard SCMI channel.



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MCTP standardizes SCMI channel transport

- SCMI defines the channel abstraction, but not a common transport binding
- Monolithic SoCs often implement channels through dedicated memory access
- Multi-vendor chiplets need a standardized, routable transport
- DMTF MCTP is the selected transport

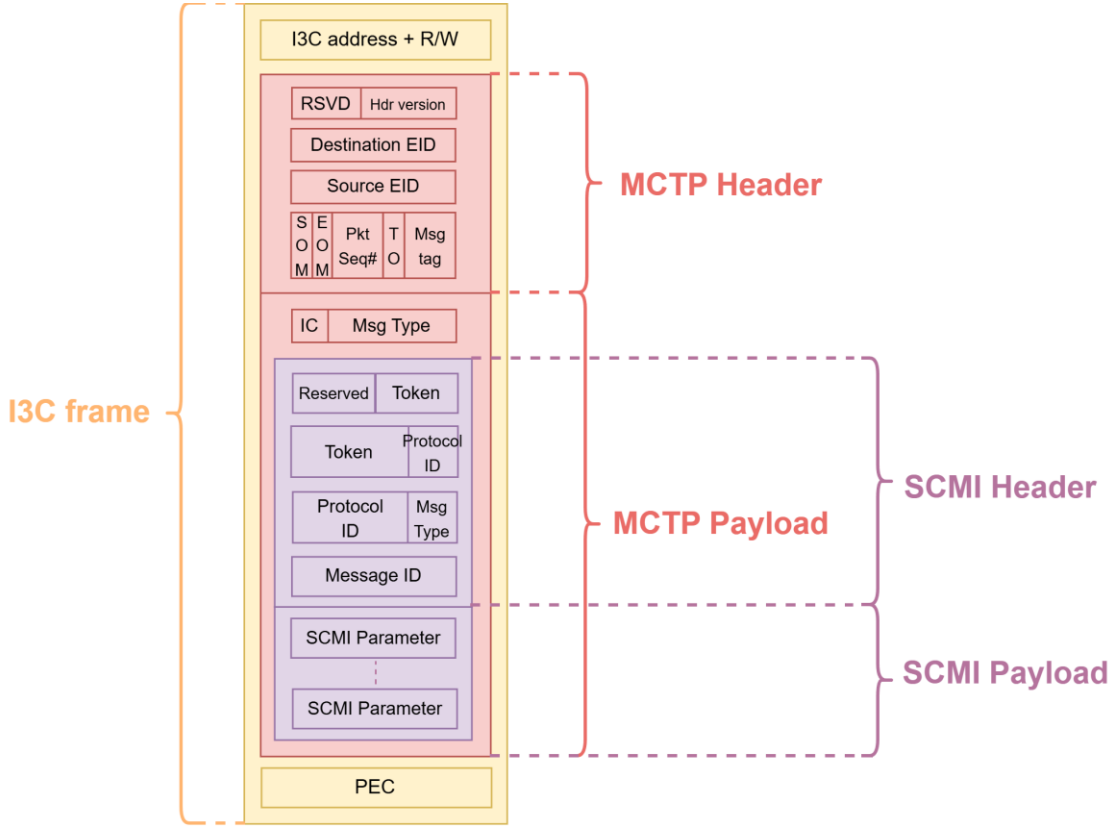


MCTP routing example

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MCTP maps SCMI across multiple physical layers

- MCTP supports mappings to multiple physical layers
- The SCMI-over-I3C mapping is defined
- Mappings for UCIe and OCP BoW are in development



SCMI over I3C framing

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Current group activity – 3 specs are ongoing

SCMI Over I3C

- **Status:** In final review
- The purpose of the **CMI-SMI3CP specification** is to define a standard way of transporting SCMI messages over I3C System Control interfaces in a chiplet-based system by leveraging the MCTP over I3C transport binding specification.

SCMI Over UCIe

- **Status:** Initial draft
- The purpose of the **CMI-SM-UCIe specification** is to define a standard way of transporting SCMI messages over the UCIe management path in a chiplet-based system, using the MTP and UMAP protocols already defined by UCIe.

SCMI Over BoW

- **Status:** Not started
- The purpose of the **CMI-SM-BoW specification** is to define a standard way of transporting SCMI messages over OCP BoW and OCP Link interfaces in a chiplet-based system.

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Call to Action

- Help advance chiplet interoperability beyond the physical layer
- **Get involved**
  - Weekly meeting · 1:00 PM PST
    - Join the weekly meeting
- **Status:** SCMI over I3C v0.93 is in review
- **Next milestone:** SCMI over UCIe and OCP BoW targeted for early 2027
- **Resources**
  - Project wiki · OpenCompute Chiplet Systems
  - SCMI over I3C · v0.92 public draft

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Thank You!

